

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE

PHYSICS

Paper 1

Wednesday 8 November 2023 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the Physics Equations Sheet (enclosed).

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information

- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



N 0 V 2 3 9 2 0 3 1 0 1

Answer **all** questions in the spaces provided.

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0 1

Different energy resources can be used to generate electricity.

0 1 . 1

Some energy resources are renewable energy resources.

What is meant by 'renewable energy resource'?

[1 mark]

Tick (✓) **one** box.

An energy resource that can be replaced as quickly as it is used.

☐

An energy resource that does not harm the environment.

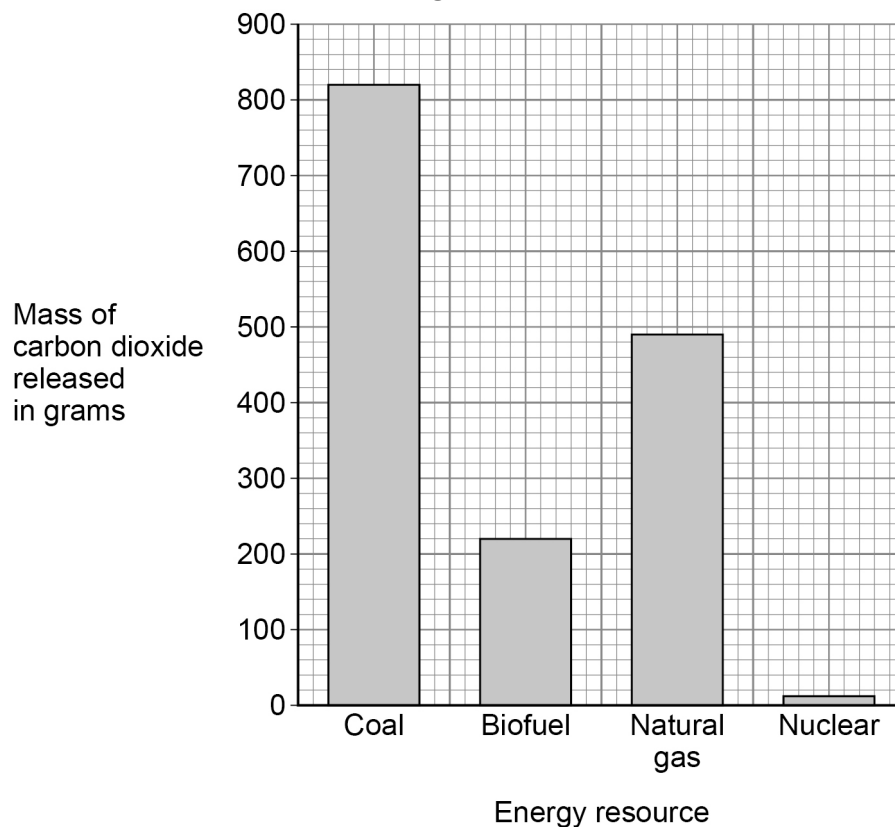
☐

An energy resource that does not produce carbon dioxide.

☐

Figure 1 shows the mass of carbon dioxide released in grams when 1 kWh of electricity is generated using different energy resources.

Figure 1



0 1 . 2 Which energy resource in **Figure 1** is renewable?

[1 mark]

Tick (✓) **one** box.

Coal

☐

Biofuel

☐

Natural gas

☐

Nuclear

☐

0 1 . 3 Which energy resource in **Figure 1** contributes the most to global warming?

[1 mark]

Tick (✓) **one** box.

Coal

☐

Biofuel

☐

Natural gas

☐

Nuclear

☐

0 1 . 4 Determine the mass of carbon dioxide produced when biofuel is used to generate 40 kW h of electricity.

Use **Figure 1**.

Give your answer in grams.

[3 marks]

Mass = _____ g

Turn over ►



The electricity generated is transferred to houses to be used by electrical appliances.
A 2.0 kW heater is switched on for 6.0 hours.

01.5 Calculate the energy transferred by the heater.

Use the Physics Equations Sheet.

Give your answer in kWh.

[2 marks]

Energy = _____ kWh

01.6 Calculate the cost of using the heater.

cost of 1 kWh = \$0.35

Use your answer from Question **01.5**.

[2 marks]

Cost = \$ _____

10



0 2

A force is a vector quantity.

0 2 . 1

What is meant by 'a vector quantity'?

[1 mark]Tick (✓) **one** box.

A quantity that has a direction only.

☐

A quantity that has magnitude and a direction.

☐

A quantity that has magnitude only.

☐**0 2 . 2**

Which two of the following are vector quantities?

[2 marks]Tick (✓) **two** boxes.

Distance

☐

Energy

☐

Momentum

☐

Temperature

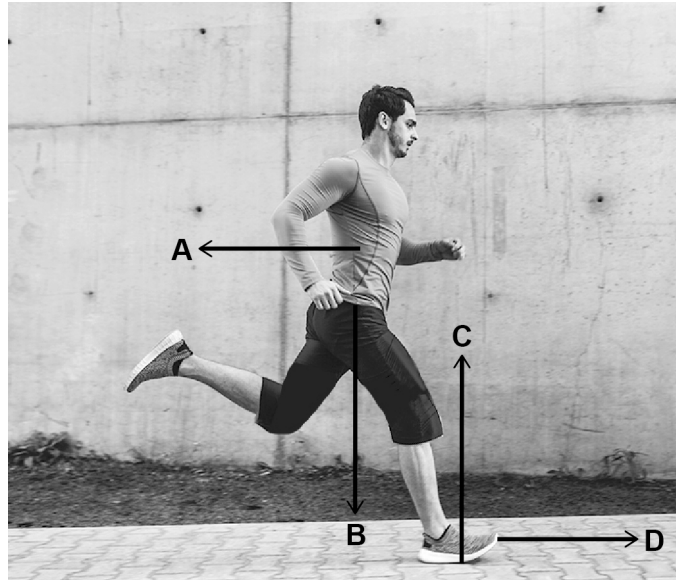
☐

Velocity

☐**Question 2 continues on the next page****Turn over ►**

Figure 2 shows an athlete running. The arrows represent the forces acting on the athlete.

Figure 2



0 2 . 3 Which arrow in **Figure 2** represents the normal contact force?

[1 mark]

Tick (✓) **one** box.

A ☐	B ☐	C ☐	D ☐
----------------------------	----------------------------	----------------------------	----------------------------

0 2 . 4 Which arrow in **Figure 2** represents air resistance?

[1 mark]

Tick (✓) **one** box.

A ☐	B ☐	C ☐	D ☐
----------------------------	----------------------------	----------------------------	----------------------------



Figure 3 shows athletes taking part in the steepest running race in Europe.

Figure 3



0 2 . 5 One athlete gains 171 500 J of gravitational potential energy during the race.
The vertical height the athlete moves through during the race is 250 m.

Calculate the mass of the athlete.

gravitational field strength = 9.8 N/kg

Use the Physics Equations Sheet.

[3 marks]

Mass = _____ kg

Question 2 continues on the next page

Turn over ►



0	2	.	6
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The athlete has an average speed of 1.25 m/s during the race.

Determine the time taken for the athlete to run a distance of 400 m.

Use the Physics Equations Sheet.

[3 marks]

Time taken = _____ s

11



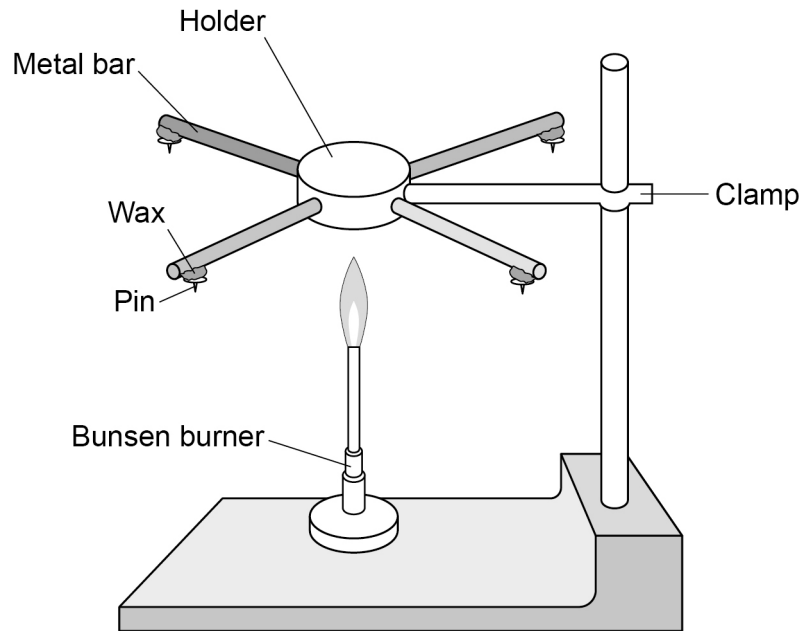
0 3

A student had four metal bars. Each bar was a different metal.

The student investigated which metal was the best thermal conductor.

Figure 4 shows some of the equipment used by the student.

Figure 4



A pin was attached with wax to the end of each bar.

The student used a Bunsen burner to heat the holder where the bars were joined.

When the wax on a bar melted, the pin fell.

The student recorded the time taken for the wax on each bar to melt.

0 3 . 1

Give **two** control variables for this investigation.

[2 marks]

- 1 _____
- 2 _____

Question 3 continues on the next page

Turn over ►



The student completed the investigation and calculated the mean time taken for the wax on each bar to melt.

Table 1 shows the results.

Table 1

Metal	Time taken to melt the wax in seconds			
	Test 1	Test 2	Test 3	Mean
Aluminium	83	85	87	85
Copper	68	65	62	65
Iron	196	195	197	196
Zinc	112	115	118	115

0 3 . 2 Which metal in **Table 1** has the most precise results in this investigation?

Give a reason for your answer.

[2 marks]

Metal _____

Reason _____

0 3 . 3 Determine the uncertainty in the results for copper in **Table 1**.

[2 marks]

Uncertainty = $\pm$ _____ s

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 One end of each metal bar is heated.

Explain how energy is transferred through a metal bar by conduction.

[3 marks]

9

Turn over for the next question

Turn over ►

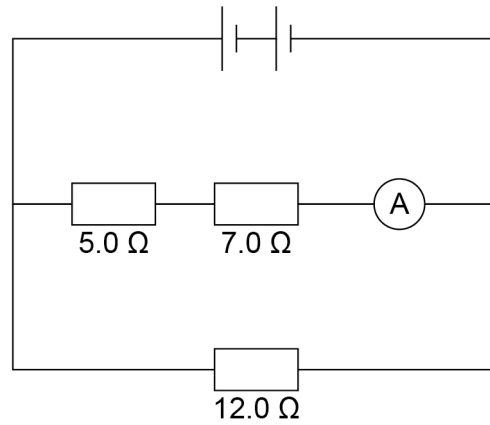


0 4

A student connected a circuit using two 1.5 V cells, an ammeter and some resistors.

Figure 5 shows the circuit.

Figure 5

**0 4 . 1**

What is the total potential difference across the cells in **Figure 5**?

[1 mark]

Tick (✓) **one** box.

1.5 V

☐

3.0 V

☐

6.0 V

☐

12.0 V

☐

0 4 . 2 The reading on the ammeter in **Figure 5** on page 12 is 0.25 A.

What is the current in the $5.0\ \Omega$ resistor in **Figure 5**?

[1 mark]

Tick (✓) **one** box.

0.08 A

☐

0.10 A

☐

0.25 A

☐

0.50 A

☐

0 4 . 3 Determine the potential difference across the $5.0\ \Omega$ resistor in **Figure 5**.

Use your answer from Question **04.2** and the Physics Equations Sheet.

[2 marks]

Potential difference = _____ V

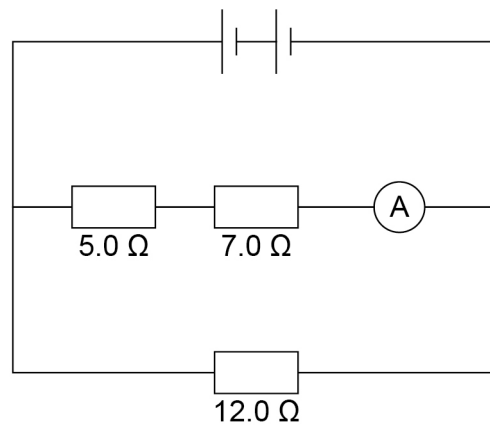
Question 4 continues on the next page

Turn over ►



Figure 5 is repeated below.

Figure 5



0 4 . 4 The total current through the cells in the circuit in **Figure 5** is 0.50 A.

Calculate the charge flow through the cells in the circuit in 22 seconds.

Use the Physics Equations Sheet.

Give the unit.

[4 marks]

Charge = _____ Unit _____

0 4 . 5 What is the total resistance of the resistors in the circuit in **Figure 5**?

[1 mark]

Tick (✓) **one** box.

Less than $12.0\ \Omega$

☐

Equal to $12.0\ \Omega$

☐

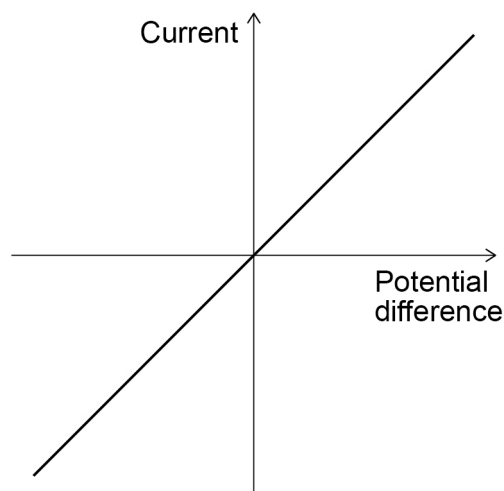
More than $12.0\ \Omega$

☐


0	4	.	6
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Figure 6 shows a sketch graph of the relationship between current and potential difference for a resistor at constant temperature.

Figure 6



Explain the shape of the sketch graph in **Figure 6**.

[3 marks]

12

Turn over for the next question

Turn over ►



0 5

A student investigated how the force applied to a spring affects the extension of the spring.

0 5 . 1

Draw **one** line from each type of variable to the variable in this investigation.

[2 marks]**Type of variable**

Independent variable

Dependent variable

Variable

Extension of the spring

Force applied to the spring

Mass of the spring

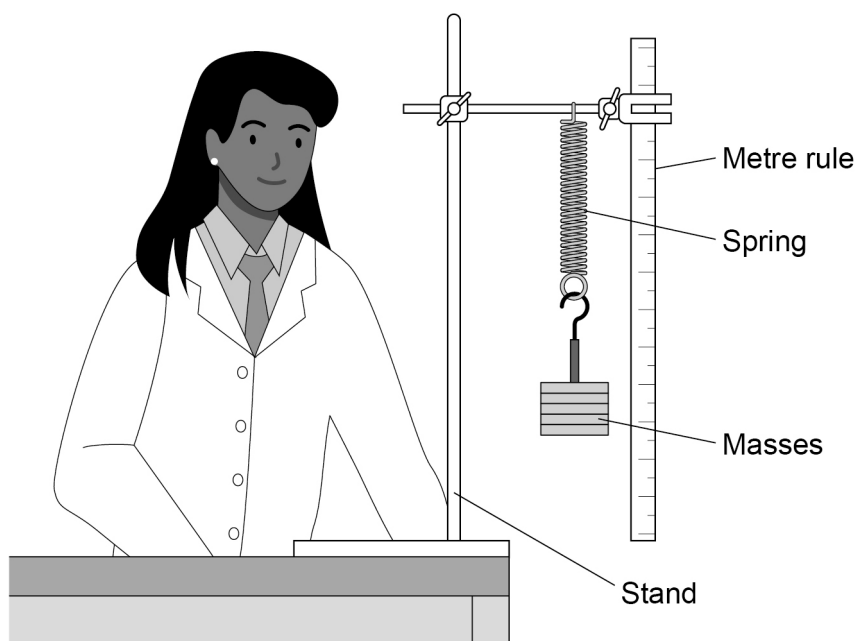
Original length of spring

Spring constant of the spring



Figure 7 shows the equipment used.

Figure 7



0 5 . 2 Identify **two** hazards with this investigation.

Suggest a safety precaution for each hazard.

[4 marks]

Hazard _____

Safety precaution _____

Hazard _____

Safety precaution _____

Question 5 continues on the next page

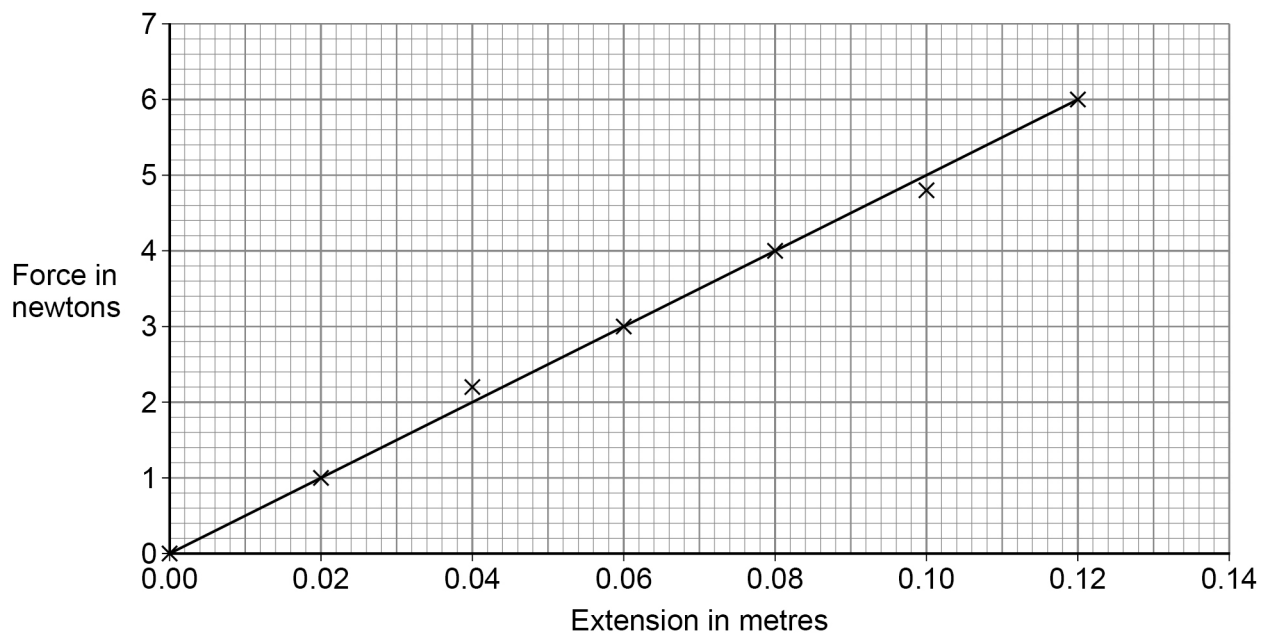
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Figure 8 shows the results.

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Figure 8



0 5 . 3 Determine the spring constant of the spring.

Use **Figure 8**.

[2 marks]

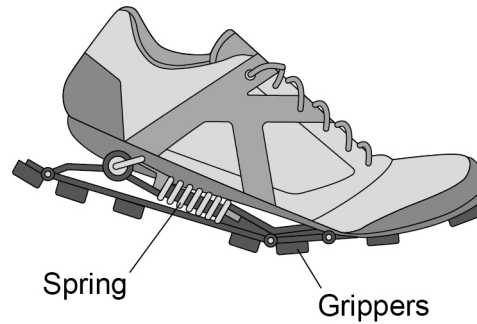
Spring constant = _____ N/m



Figure 9 shows a shoe used by runners.

The shoe has a spring attached. The spring reduces the force exerted on the runner's foot.

Figure 9



When the runner's foot pushes down on the ground the spring compresses.

0 5 . 4

The required compression of the spring is 0.050 m to gain maximum benefit from the shoes.

Different springs are used depending on the mass of the runner.

A runner with a greater mass would need a spring with a different spring constant from a runner with a smaller mass.

Explain why.

[3 marks]

Question 5 continues on the next page

Turn over ►



0	5	.	5
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As the runner's shoe is pushed down on the ground the spring compresses.

Explain how the compression of the spring reduces the force on the runner's foot.

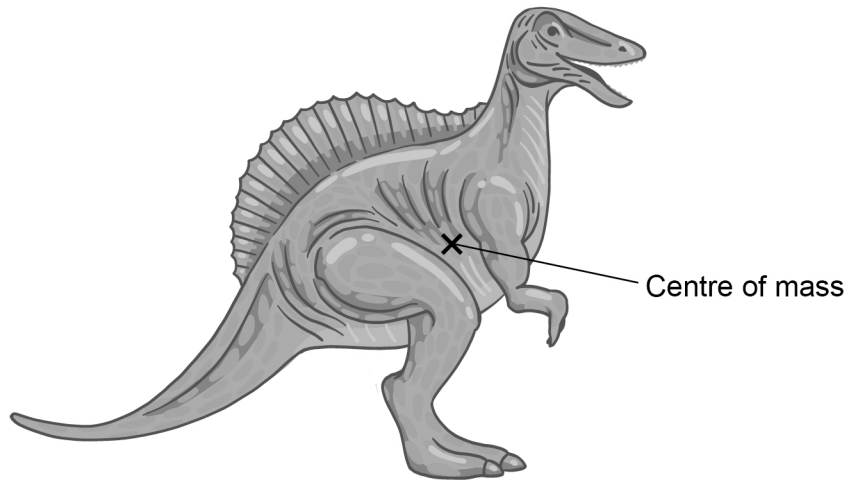
Use ideas about **momentum** in your answer.

[3 marks]

14



0 6

Figure 10 shows a toy.The centre of mass of the toy is marked **X**.**Figure 10**

0 6

. 1

Draw the line of action of the weight of the toy on **Figure 10**.**[1 mark]**

0 6

. 2

What is meant by 'centre of mass'?

[1 mark]

0 6

. 3

Give **two** ways the design of the toy could be changed to increase its stability.**[2 marks]**

1

2

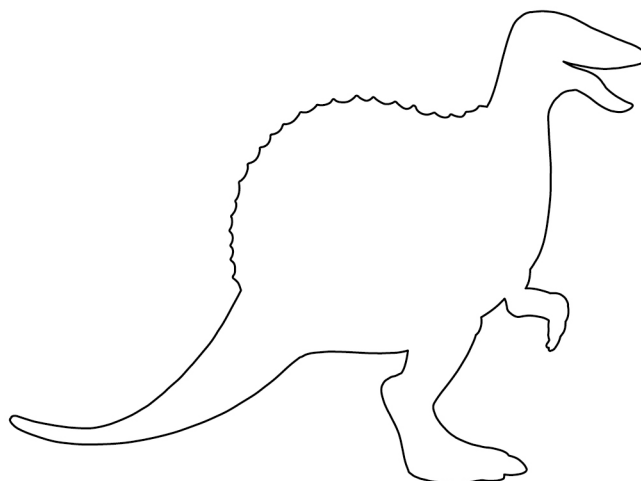
Question 6 continues on the next page**Turn over ►**

0 6 . 4

A picture of the toy was cut out of a piece of card.

Figure 11 shows the toy-shaped piece of card.

Figure 11



Describe how a student could find the centre of mass of the toy-shaped piece of card.

You may include a diagram.

[4 marks]



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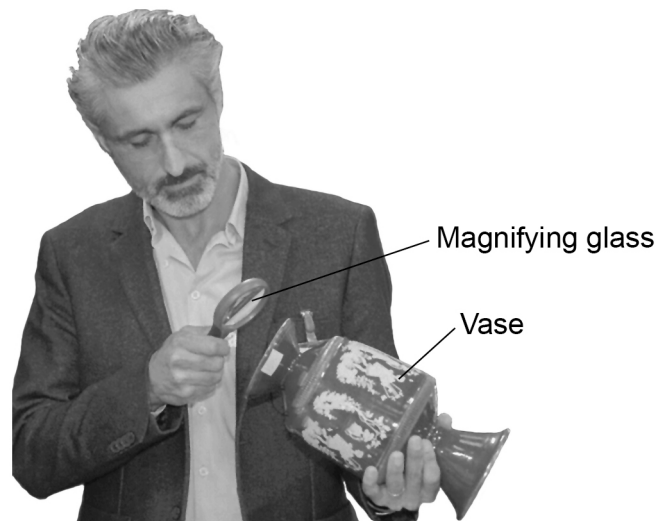
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07

Figure 12 shows a person looking through a magnifying glass at the bottom of a vase.

Figure 12



07.1

Explain what happens to light rays incident on the glass surface of the magnifying glass.

[2 marks]

07.2

What type of lens is used in a magnifying glass?

[1 mark]

07.3

The magnifying glass produces an image. The image is magnified.

Give **two** other properties of the image.

[2 marks]

1

2



0	7	.	4
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The person in **Figure 12** is looking at some symbols on the bottom of the vase.

When looking through the magnifying glass, the height of one of the symbols appears to be 24 mm.

The magnification is 2.5

Calculate the actual height of the symbol.

Give your answer in millimetres.

Use the Physics Equations Sheet.

[3 marks]

Actual height of symbol = _____ mm

Question 7 continues on the next page

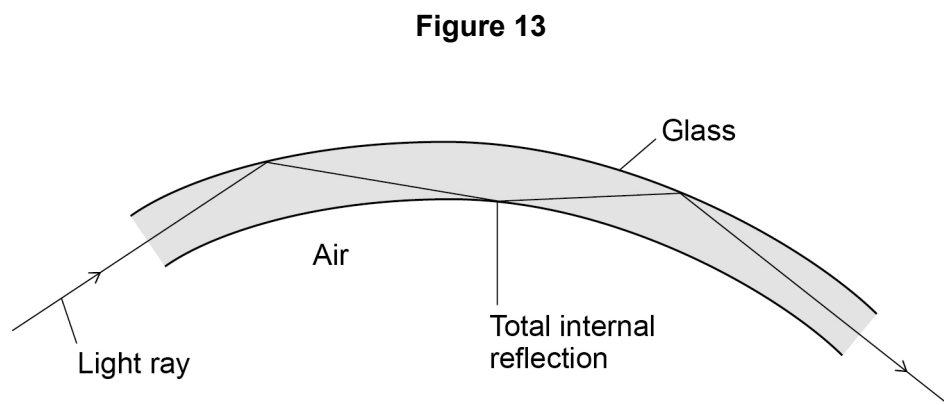
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The person in **Figure 12** on page 24 uses an optical fibre to look inside the vase.

Light travels through an optical fibre using total internal reflection.

Figure 13 shows total internal reflection in an optical fibre.



0 7 . 5

Give **two** conditions needed for light to totally internally reflect.

[2 marks]

1 _____

2 _____



A ray of light is incident on the end of the optical fibre. The angle of incidence is 35° and the angle of refraction is 24°

Use the Physics Equations Sheet.

[5 marks]

[illegible]

Critical angle = _____ °

15

Turn over for the next question

Turn over ►



0 8

The radioactive isotope strontium-89 can be represented by the symbol:



0 8 . 1

There are **two** types of particles in a nucleus of an atom.

Give the name **and** number of each type of particle in a nucleus of strontium-89.

[2 marks]

Particle 1 _____

Particle 2 _____

Strontium-89 emits a beta particle to become yttrium (Y).

0 8 . 2

Describe how a beta particle is emitted from a nucleus of strontium-89.

[2 marks]

0 8 . 3

Complete the nuclear equation for the decay of strontium-89.

[2 marks]



08.4

A sample of strontium-89 has a count rate of 192 counts per second.

The half-life of strontium-89 is 50.5 days.

Calculate the count rate of the sample of strontium-89 after 202 days.

[3 marks]

Count rate after 202 days = _____ counts per second

08.5

Ionising radiation from strontium-89 is used in the treatment of cancer.

Explain why ionising radiation is used in the treatment of cancer even though the radiation damages healthy cells.

[2 marks]

11

END OF QUESTIONS

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